Indeed, denote the latter by J. Then J is a nilpotent ideal in $A \otimes B$, as $\operatorname{Rad}(A)$ and $\operatorname{Rad}(B)$ are nilpotent. On the other hand, $(A \otimes B)/J = (A/\operatorname{Rad}(A)) \otimes (B/\operatorname{Rad}(B))$, which is a product of two semisimple algebras, hence semisimple. This implies $J \supset \operatorname{Rad}(A \otimes B)$. Altogether, by Proposition 3.5.3, we see that $J = \operatorname{Rad}(A \otimes B)$, proving the claim.

Thus, we see that

$$(A \otimes B)/\operatorname{Rad}(A \otimes B) = A/\operatorname{Rad}(A) \otimes B/\operatorname{Rad}(B).$$

Now, M is an irreducible representation of $(A \otimes B)/\operatorname{Rad}(A \otimes B)$, so it is clearly of the form $M = V \otimes W$, where V is an irreducible representation of $A/\operatorname{Rad}(A)$ and W is an irreducible representation of $B/\operatorname{Rad}(B)$. Also, V, W are uniquely determined by M (as all of the algebras involved are direct sums of matrix algebras). $\square$